Technische Universität München

Mathematics preparatory course

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6 Primes and divisibility

6.1 Prime factorization

Recall that $|$ denote the "divides relation" on $\mathbb{N}$, i.e., $a|b$ has to be read as "a is a divisor of b" or (equivalently) as "b is a multiple of a".

Formally: $a|b$:iff $b = ka$ for some $k \in \mathbb{N}$.

A positive integer $n \in \mathbb{N}$ is called a *prime* :iff $n > 1$ with $\{d \in \mathbb{N} : d|n\} = \{1, n\}$.

1. Determine all prime numbers in $\{1, 2, \ldots, 50\}$.

   Use: (i) if p is prime, all of its multiples $2p, 3p, 4p, \ldots$ are certainly not prime, and (ii) if $n > 1$ is not a prime, then it has a divisor $d|n$ with $d \leq \sqrt{n}$ (why?).

2. The fundamental theorem of arithmetic is usual used without proof in school: it states that every positive integers $n \in \mathbb{N}$ can only be factorized in one unique way (up to commutativity, i.e., except for the order of the prime factors) into a product of prime numbers. Due to the uniqueness, we can compute the prime factorization of some $n > 1$ by trying to find some divisor $d|n$ and then factorize recursively d and $\frac{n}{d}$.

   E.g. $110 = 10 \cdot 11$ and $10 = 2 \cdot 5$, thus $110 = 2 \cdot 5 \cdot 11$.

   Determine the prime factorization of each of the following numbers:

   $$111, \quad 112, \quad 113, \quad 114, \quad 115$$

3. The *greatest common divisor* of $a, b \in \mathbb{N}$ is (as the name says) $\mathsf{gcd}(a, b) := \max\{d \in \mathbb{N} : d|a \text{ and } d|b\}$.

   And the *least common multiple* is $\mathsf{lcm}(a, b) := \min\{m \in \mathbb{N} : a|m \text{ and } b|m\}$.

   Both are well-defined as can be argued by means of the uniqueness of the prime factorization (why?).

   Compute the gcd and lcm for each pair of numbers listed in (c) including 110.

   (Recall that $\mathsf{lcm}(a, b) \cdot \mathsf{gcd}(a, b) = ab$.)

Now determine for each pair of numbers their least common multiple (LCM) as well as their greatest common divisor (GCD).

6.2 Number of Prime Divisors

Why does every $n \in \mathbb{N}$ have at most $\log_2(n)$ (distinct) prime divisors?

7 Rational and real numbers, fields, linear equations

7.1 Inequalities

Determine all solutions of the following inequality (over $\mathbb{R}$):

$$\frac{x-8}{x+1} < \frac{1-x}{x+3}$$

Remark: $2x^2 - 5x - 25 = (x-5)(2x+5)$.

7.2 Systems of Linear Equations

There are different algorithms for solving systems of linear equations that work for any field, e.g. "substitution" and "Gaussian elimination".

(a) *By Substitution:*
Solve the following system of linear equations over the real numbers step-by-step by solving one equation for one variable and then substituting it into another equation as shown in the lecture slides:

$$x - 2y = 4, \quad -y - z = -1, \quad -x + 3z + y = -1$$

(b) (*) *By Gaussian elimination (elementary transformations):*
Recall: if we (i) multiple a single equation by a non-zero number or (ii) add a multiple of one equation to another (distinct) equation, then the set of possible solutions does not change (as we can reverse these transformations).

For instance, consider:

$$\begin{array}{lcll} -x + 4y & = & 2 & (1) \\ 2x - y & = & 3 & (2) \end{array}$$

Transforming the equations step-by-step:

$-x + 4y$	$=$	2	(1)
$2x - y$	$=$	3	(2)
$-2x + 8y$	$=$	4	$(3) := 2 \cdot (1)$
$7y$	$=$	7	$(4) := (2) + (3)$
y	$=$	1	$(5) := (4) \cdot \frac{1}{7}$
$2x$	$=$	4	$(6) := (2) + (5)$
x	$=$	2	$(7) := (6) \cdot \frac{1}{2}$

(You may also combine the steps (3) and (4).)

Solve the following system of equations (same as in (a)) now using Gaussian elimination.

$$x - 2y = 4, \quad -y - z = -1, \quad -x + 3z + y = -1$$

(Solving systems of linear equations is in fact an important algorithmic problem, in particular, when imprecisions due to rounding have to be minimized.)

(c) More systems to solve (use either of the two methods, whatever you prefer/know from school):

(a) $2x + 6y = 9, 4x + 9y = 15$

(b) $3x + 2y = 6, 6x + 4y = 6$

(c) $3x + 4y + 5z = 2, 6x + 7z = 5$

(d) $3x + 4y + 5z = 2, 6x + 7z = 5, 2x - 3y = 3$

Also recall how that a linear equation in two variables describes a line in $\mathbb{R}^2$, while a linear equation in three variables describes a plane in $\mathbb{R}^3$. Recall that the set of all possible solutions of a system of linear equations can be understood as the intersection of the corresponding lines resp. planes, and thus the set of solutions is either empty, a point, a line, or a plane itself (assuming at most 3 variables).

Use geogebra to visualize the systems: e.g., you can simply enter the two equations $2x+6y=9$ and $4x+9y=15$ independently on the left ("Input ..."). For three variables, you need to switch from "Graphing" to "3d Calculator" (drop-down box above the drawing area).

7.3 Properties of Fields

Let F be a field with addition + and multiplication $\cdot$.

(Simply think of F as $\mathbb{Q}$ or $\mathbb{R}$.)

Show, that the following statements are always true, arguing only with the axioms for fields.

(See the slides for the field axioms.)

(a) For all $x, y, z \in F$: If $x \cdot y = 1$ and $x \cdot z = 1$, then $y = z$.

(b) For all $x \in F$: $\left(x^{-1}\right)^{-1} = x$.

(c) For all $x, y \in F$: $(x \cdot y)^{-1} = x^{-1} \cdot y^{-1}$.

(d) For all $x \in F$: $x \cdot 0 = 0$

(e) For all $x, y \in F$: If $x \cdot y = 0$, then $x = 0$ or $y = 0$.

7.4 Field Axioms for important Sets

Determine for each of the sets $\mathbb{N}$, $\mathbb{N}_0$, $\mathbb{Z}$ and $\mathbb{Q}$ with the usual addition and multiplication, which of the requirements for a field are met and which are not.

7.5 Finite Fields (*)

A non-empty set K, on which two operations $+ : F \times F \to F$ (addition) and $\cdot : F \times F \to F$ (multiplication) are defined, is called a **field**, if all of the following properties are satisfied:

Commutativity of $+$ and $\cdot$	for all $x, y \in F$: $x + y = y + x$ and $x \cdot y = y \cdot x$
Associativity of $+$ and $\cdot$	for all $x, y, z \in F$: $x + (y + z) = (x + y) + z$ and $x \cdot (y \cdot z) = (x \cdot y) \cdot z$
Neutral Elements for $+$ and $\cdot$	exists $0, 1 \in F : 0 \neq 1$ and for all $x \in K$: $x + 0 = x = 0 + x$ and $x \cdot 1 = x = 1 \cdot x$
Inverse Elements for $+$ and $\cdot$	for all $x \in F$: exists $y \in K$: $x + y = 0$ (for this y one also writes $-x$) and for all $x \in F \setminus \{0\}$: exists $y \in K \setminus \{0\}$: $xc \cdot y = 1$ (for this y one also writes x^{-1})
Distributivity of $\cdot$ over $+$	for all $x, y, z \in F$: $x \cdot (y + z) = (x \cdot y) + (x \cdot z)$

These are the usual "rules" for computing with rational and real numbers.

But there are more fields than $\mathbb{Q}$ and $\mathbb{R}$. In particular *finite fields* play an important role not only in mathematics, but also in computer science (cryptography, error correction, hashing, ...).

(a) Define an addition $(+)$ and a multiplication $(\cdot)$ on the set $\mathbb{Z}_2 := \{0, 1\}$, so that all the above requirements are satisfied.

Simply define the addition by defining the result of all possible inputs by means of the following table:

$+$	0	1
0	$\ldots$	$\ldots$
1	$\ldots$	$\ldots$

Similarly define the multiplication on $\mathbb{Z}_2$.

$\cdot$	0	1
0	$\ldots$	$\ldots$
1	$\ldots$	$\ldots$

(These tables are also called "Cayley tables".)

Note: Remember, that the results of any addition and multiplication must be an element of $\mathbb{Z}_2$ again (addition and multiplication must be "closed"). So e.g. defining $1 + 1 := 2$ is not admissible.

(b) Check, that $\mathbb{Z}_3 := \{0, 1, 2\}$ is a field w.r.t. the addition $x +_3 y := (x + y) \bmod 3$ and the multiplication $x \cdot_3 y := (x \cdot y) \bmod 3$.

Reminder: $x \bmod n$ denotes the remainder when dividing $x \in \mathbb{Z}$ by $n \in \mathbb{N}$ – the standard convention is that the remainder is a nonnegative integer in $\mathbb{Z}_n := \{0, 1, \ldots, n-1\}$. So $7 \bmod 3 = 1$, but also $-2 \bmod 3 = 1$.

Checking commutativity and the existence of the required inverse elements, can be done using the tables.

Checking associativity can become a bit annoying; there are (at least) two possible approaches:

- Brute force:

  As there are only three values, you can simply check for all combinations of $x, y, z \in \{0, 1, 2\}$ whether all properties are satisfied.

  (If you know how to program, use a computer; that's what they were invented for originally.)

- Math:

  $(x + y) \bmod n = x + y - kn$ for some $k \in \mathbb{Z}$ (namely: $k = \lfloor \frac{x+y}{n} \rfloor$, but this is unimportant).

  Hence: $(((x+y) \bmod n)+y) \bmod n = (x+y+z-kn) \bmod n = x+y+z+-kn-ln = x + y + z - (k + l)n = (x + y + z) \bmod n$.

(c) In general, if p is a prime number, then $\mathbb{Z}_p = \{0, 1, \ldots, p-1\}$ becomes a field w.r.t. addition and multiplication modulo p.

Try to solve the following systems of equations over the field $\mathbb{Z}_5$ (with its addition $+_5$ and multiplication $\cdot_5$):

$$(2 \cdot_5 x) +_5 (2 \cdot_5 y) = 0, \quad x +_5 (4 \cdot_5 y) = 2$$

The two approachs for solving systems of linear equations discussed in Ex 7.2 actually do not require any specific property of $\mathbb{Q}$ or $\mathbb{R}$, only the generic properties of a field are required. The one important thing to remember is that the inverse elements w.r.t. addition and multiplication depend of course on the specific addition and multiplication used, so e.g. 3^{-1} modulo 5 is not 0.3333.... but 2 as $(3 \cdot 2) \bmod 5 = 6 \bmod 5 = 1$.

Remark: If n is a composite, then $\mathbb{Z}_n$ w.r.t. addition and multiplication modulo n is never a field. E.g. consider $n = 6$. Then we have $(2 \cdot 3) \bmod 6 = 0$, but in every field it has to hold that, if $a \cdot b = 0$, then $a = 0$ or $b = 0$. (See Ex 7.4.)

(d) Show that for every finite field every row and column of the Cayley tables of its addition and multiplication must contain every element exactly once (i.e. they are permutations of the field elements).

(e) Can you define an addition and multiplication on $\mathbb{Z}_4 = \{0, 1, 2, 3\}$ so that all properties of a field are satisfied?

Hint: Computing modulo 4 does not work here. Start by defining $0 = x + x$ for all $x \in \mathbb{Z}_4$, and choose $0, 1$ to be the neutral element wrt. addition and multiplication, respectively (as we are all used to).